AI Model Card

MRCAT Prostate + Auto-Contouring

Company: Philips | Generated: 10/23/2026

General Information

Product Name:	MRCAT Prostate + Auto-Contouring
Company:	Philips
Version:	N/A
Category:	Image Synthesis
Secondary Categories:	Auto-Contouring
Release Date:	3/31/2016
Last Updated:	8/25/2026
Description:	MRCAT Prostate is indicated for radiotherapy treatment planning for prostate cancer patients; radiotherapy target volumes for patients with prostate cancer, including anatomies in the pelvic region and lymph nodes with margins, up to the L3 vertebra, are suitable for MR-only simulation with MRCAT Prostate. (Source: Philips Ingenia MR-RT Instructions for Use, Release RTgo 5.12, 3000 113 93922/781, 2024-06, pp. 10–11. IFU URL: https://www.documents.philips.com/assets/Instruction%20for%20Use/20250625/aecaea1f0eb749a7babfb30700bf34b8.pdf?feed=ifu_docs_feed, retrieved 2026-06-15, publicly accessible.)

Evidence Classification

Rigor E2 — External, independent or multi-center validation **Efficacy L4 — Therapeutic/patient management efficacy**

Readiness R3 — Enhanced implementation readiness

+ Vendor-independent **+ Multi-centre** **+ Multi-national** **- Prospective** **+ External validation**

15 scored publication(s) · 15 listed · score origin: papers
Rigor: Aggregate across modules. Image Synthesis (MRCAT): E2 — independent peer-reviewed dosimetric validation studies directly on MRCAT prostate (Tyagi 2017 DOI aa5452, Kemppainen 2017, Maspero 2017 DOI aa4fe7 photon, Maspero 2017 DOI aa9677 proton). MESCAL 2026 (Cusumano, Maspero et al. Radiother Oncol, DOI 10.1016/j.radonc.2026.111530) included as indirect-comparative community benchmark. Persson 2017 MR-OPERA removed 2026-06-15 — it validates Spectronic MRiPlanner, not Philips MRCAT. Auto-Contouring: E0 — no module-specific peer-reviewed publications. See categoryEvidence for per-module detail. Christiansen 2017 removed 2026-08-25 — the cited DOI resolves to an unrelated paper and no Christiansen MRCAT publication exists in Europe PMC.
Efficacy: Aggregate. MRCAT enables MR-only prostate workflows in clinical use at several centers (I1). Auto-Contouring component has no published clinical impact data (I0). 2026-08-25 Wave 5: clinical impact raised I1 !' I2, (doi 10.1186/s13014-017-0854-4), which reports a routine MR-only prostate simulation and planning workflow in 48 patients with process-level endpoints.

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Readiness: MRCAT MR-only prostate workflow is in routine clinical use at multiple centers with peer-reviewed dosimetric validation (including photon and proton MR-only prostate, Maspero 2017a/b); auto-contouring module less well characterised.

Key Features

- Synthetic CT (MRCAT) generation for MR-only workflow
- Model-based adaptive auto-contouring
- MR-only workflow (eliminates CT)
- One-click automated workflow
- Bulk motion correction via bone registration
- < 5 minute processing time

Technical Specifications

Input Formats:	DICOM
Output Formats:	DICOM CT; DICOM-RTSTRUCT
Processing Time:	< 5 minutes (parallel with scanning)
Integration Options:	Ingenia MR-RT console; DICOM export to any TPS
Deployment Options:	On-premises
Target Population:	Adult patients (accuracy: average distance < 1.5mm)
Modality Support:	MRI
Target Anatomy:	Pelvis
Disease Targeted:	Prostate Cancer

Supported Structures (9)

Pelvis — 9

Prostate (anatomical)	Seminal Vesicles	Bladder (inner wall)
Bladder (outer wall)	Rectum	Penile Bulb
Femoral Head (L)	Femoral Head (R)	Body Outline

Evidence & Limitations

Clinical Evidence:	N/A
Limitations:	Patient selection (IFU): not suitable for patients with large metal objects (e.g. hip prosthesis) in the imaging volume, cancers other than prostate cancer, bone anomalies/diseases in the pelvic area, or body diameter in the pelvic area exceeding 50 cm (L-R) or 30 cm (A-P) within the planning FOV.; General Ingenia MR-RT exclusions: MRI contraindications, MR contrast-agent contraindications, claustrophobia, inability to tolerate position/scan time, treatment position unsuitable for

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Limitations (cont.):

MRI, patient weight > 250 kg.; Operational: MRCAT Prostate assigns discrete (bulk) HU values to tissue classes — images may not be suitable for soft-tissue dose evaluation in the very low-dose region.; Operational: signal-void volumes inside the body outline other than compact bone are interpreted as water-rich or fatty tissue, which can mis-categorize rectal gas, foreign material, or implants and locally affect dose calculation accuracy.; Operational: generate MRCAT Prostate images before any contrast agent is administered to the patient — otherwise the MRCAT post-processing step may fail.; Operational: MRCAT must not be used with restricted dB/dt or with a gradient slew rate restricted below the MR system limit; MRCAT images must not be post-processed.; Source: Philips Ingenia MR-RT IFU, RTgo 5.12 (2024-06), pp. 10–11 and 63–68. URL: https://www.documents.philips.com/assets/Instruction%20for%20Use/20250625/aecaea1f0eb749a7babfb30700bf34b8.pdf?feed=ifu_docs_feed. Retrieved 2026-06-15.

Evidence by Source (15)

A study of Bayesian deep network uncertainty and its application to synthetic radiotherapy treatment planning

Law MW et al. · Medical Physics · 2023 · DOI 10.1002/mp.16666

E1 L2

Wave 5A flagged paper, routed 2026-08-25: 59 prostate-cancer patients with same-day pelvic CT and MR; MRCAT is the commercial atlas-based comparator (MAE 54.87 HU, 2%/2 mm gamma 98.56%) against a proposed Bayesian network. MRCAT is genuinely evaluated, but as comparator.

https://doi.org/10.1002/mp.16666

Dosimetric and workflow evaluation of first commercial synthetic CT software for clinical use in pelvis

Tyagi N, Fontenla S, Zhang J, et al. · Physics in Medicine & Biology · 2017 · DOI 10.1088/1361-6560/aa5452

E1 L2

Single-centre dosimetric and workflow evaluation of the first commercial (MRCAT) sCT software for pelvis.

https://doi.org/10.1088/1361-6560/aa5452

Clinical workflow for MR-only simulation and planning in prostate

Tyagi N, Fontenla S, Zelefsky M, et al. · Radiation Oncology · 2017 · DOI 10.1186/s13014-017-0854-4

E1 L4 Sets product level

Implementation of a routine MR-only prostate simulation/planning workflow in 48 patients with process and acquisition-time endpoints — workflow-level, not only geometric.

https://doi.org/10.1186/s13014-017-0854-4

Quantification of confounding factors in MRI-based dose calculations as applied to prostate IMRT

Maspero M, Seevinck PR, Schubert G, et al. · Physics in Medicine & Biology · 2017 · DOI 10.1088/1361-6560/aa4fe7

E1 L2

Single-centre framework isolating confounding factors in MRCAT-based prostate IMRT dose calculation.

https://doi.org/10.1088/1361-6560/aa4fe7

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Feasibility of MR-only proton dose calculations for prostate cancer radiotherapy using a commercial pseudo-CT generation method

Maspero M, van den Berg CAT, Landry G, et al. · Physics in Medicine & Biology · 2017 · DOI 10.1088/1361-6560/aa9677

E1 L2

First feasibility assessment of the commercial MRCAT pseudo-CT for prostate IMPT dose calculation.

https://doi.org/10.1088/1361-6560/aa9677

Magnetic resonance-only simulation and dose calculation in external beam radiation therapy: a feasibility study for pelvic cancers

Kemppainen R, Suilamo S, Tuokkola T, et al. · Acta Oncologica · 2017 · DOI 10.1080/0284186X.2017.1293290

E1 L2

Extends MRCAT beyond prostate to rectal and gynaecological pelvic cancers; supporting context for prostate.

https://doi.org/10.1080/0284186X.2017.1293290

Accuracy and precision of patient positioning for pelvic MR-only radiation therapy using digitally reconstructed radiographs

Kemppainen R, Vaara T, Joensuu T, Kiljunen T · Physics in Medicine & Biology · 2018 · DOI 10.1088/1361-6560/aaad21

E1 L2

Positioning accuracy from MRCAT-derived DRRs against CT-based reference.

https://doi.org/10.1088/1361-6560/aaad21

Geometric and dosimetric impact of anatomical changes for <scp>MR</scp> onl prostate

Nejad Davarani SP et al. · J Applied Clin Med Phys · 2019 · DOI 10.1002/acm2.12551

E1 L2

Single-centre characterisation of intra-fractional anatomical change in an MRCAT-based MR-only prostate workflow.

https://doi.org/10.1002/acm2.12551

Validation of dose distribution computation on sCT images generated from MRI scans by Philips MRCAT

Bratova I, Paluska P, Grepl J, et al. · Reports of Practical Oncology and Radiotherapy · 2019 · DOI 10.1016/j.rpor.2019.02.001

E1 L2

Independent single-centre dosimetric validation naming Philips MRCAT in the title.

https://doi.org/10.1016/j.rpor.2019.02.001

MRI-based IMPT planning for prostate cancer

Depauw N, Keyrilainen J, Suilamo S, et al. · Radiotherapy and Oncology · 2020 · DOI 10.1016/j.radonc.2019.10.010

E1 L2

Retrospective evaluation of MRCAT-derived stopping-power estimates for prostate IMPT.

https://doi.org/10.1016/j.radonc.2019.10.010

Dosimetric evaluation of MR-derived synthetic-CTs for MR-only proton treatment planning

Aramburu Núñez D et al. · Medical Dosimetry · 2020 · DOI 10.1016/j.meddos.2020.01.005

E1 L2

Proton PBS dose recalculation on MRCAT sCT for prostate; ten patients, single centre.

https://doi.org/10.1016/j.meddos.2020.01.005

Magnetic resonance image-based tomotherapy planning for prostate cancer

Jung SH, Kim J, Chung Y, et al. · Radiation Oncology Journal · 2020 · DOI 10.3857/roj.2020.00101

E1 L2

Retrospective comparison of MRCAT-based versus planning-CT helical tomotherapy plans in 16 patients.

https://doi.org/10.3857/roj.2020.00101

For information only. Not a clinical validation or endorsement; verify with the manufacturer and local commissioning.

Quality evaluation of synthetic CT using a gamma analysis for prostate cancer MR-only radiotherapy

Babka V, Rehounek L, Jakubcova I, et al. · Reports of Practical Oncology and Radiotherapy · 2025 · DOI 10.5603/rpor.105857

E1 L2

Independent single-centre HU/gamma QA methodology applied to MRCAT prostate sCT.

https://doi.org/10.5603/rpor.105857

Evaluation of MRI-derived synthetic CT in prostate cancer carbon ion radiotherapy: impact of fixation shell

Ishizawa M, Haga K, Sato H, et al. · International Journal of Particle Therapy · 2026 · DOI 10.1016/j.ijpt.2026.101298

E1 L2

Extends MRCAT sCT evaluation to carbon-ion therapy, where stopping-power sensitivity is highest.

https://doi.org/10.1016/j.ijpt.2026.101298

Standardizing MRI-only radiotherapy commissioning: Benchmark dataset and acceptance levels from the MESCAL initiative

Cusumano D, Maspero M, Vellini L, et al. · Radiotherapy and Oncology · 2026 · DOI 10.1016/j.radonc.2026.111530

E2 L2 Sets product level

Multi-centre, multi-vendor benchmark of commercial sCT algorithms with defined acceptance levels; dosimetric/image endpoints only.

https://doi.org/10.1016/j.radonc.2026.111530

Regulatory Information

CE: cleared FDA: 510k_cleared

CE Certification:	cleared
CE Details:	cleared (Class IIb)
FDA Status:	510k_cleared
FDA Details:	510k_cleared
Intended Use Statement:	MRCAT Prostate is indicated for radiotherapy treatment planning for prostate cancer patients; radiotherapy target volumes for patients with prostate cancer, including anatomies in the pelvic region and lymph nodes with margins, up to the L3 vertebra, are suitable for MR-only simulation with MRCAT Prostate. (Source: Philips Ingenia MR-RT Instructions for Use, Release RTgo 5.12, 3000 113 93922/781, 2024-06, pp. 10–11. IFU URL: https://www.documents.philips.com/assets/Instruction%20for%20Use/20250625/aecaea1f0eb749a7babfb30700bf34b8.pdf?feed=ifu_docs_feed, retrieved 2026-06-15, publicly accessible.)

Market Information

Market Presence:	2015
On Market Since:	2015
Distribution Channels:	Direct sales

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Contact Information

Website:	https://www.usa.philips.com/healthcare/product/HCNMRB780/mrcat-prostate-auto-contouring-mr-rt-clinical-application
Company URL:	https://www.philips.com/healthcare
Product URL:	https://www.usa.philips.com/healthcare/product/HCNMRB780/mrcat-prostate-auto-contouring-mr-rt-clinical-application
Logo URL:	/logos/philips.png
Logo Source:	Logo source: /logos/philips.png
Support Email:	N/A

Quality Assurance

Last Revised:	8/27/2026
Company Revision Date:	N/A
Source:	Philips product documentation (2019); releaseDate from Philips press release (31 Mar 2016); per-category evidence updated 2026-06-15 with independent peer-reviewed MRCAT prostate dosimetric validation studies (Tyagi 2017, Kemppainen 2017, Maspero 2017a photon DOI aa4fe7, Maspero 2017b proton DOI aa9677) and indirect-comparative MESCAL 2026 benchmark (Cusumano, Maspero et al. Radiother Oncol, DOI 10.1016/j.radonc.2026.111530); top-level limitations and intended-use refreshed from Philips Ingenia MR-RT IFU, RTgo 5.12, 3000 113 93922/781 (2024-06), pp. 10–11, 63–68, retrieved 2026-06-15; Wave 5 literature-expansion sweep 2026-08-25 (Europe PMC 2014-2026, alias-gated, DOIs verified against Crossref)
GitHub URL:	https://github.com/DLinRT-eu/dlinrteu-website/tree/main/src/data/products/image-synthesis/philips-mrcat-prostate.ts

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